

THROWN

Product Requirements Document — v2.0 (Deduction Rework)

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1. Product Overview

THROWN is a competitive, real-time, voting-based social deduction game set in the Volta Chronicles universe, developed and published by Super Duper Techno (SDT). Players are secretly assigned roles each round — most are Investigators, one or two are hidden Masks — and must use discussion, clue tokens, and majority voting to identify and eliminate the Mask(s) before a fixed round limit runs out.

This document supersedes the original v1.0 PRD's core gameplay loop (Rock Paper Scissors + dare/challenge system). Account, auth, friends, chat, leaderboard, notification, and technical infrastructure sections carry over from v1.0 with adaptations noted in each section.

Property	Detail
Product Name	THROWN
Studio	Super Duper Techno (SDT)
Platform	Web (PWA) — Phase 1
Target Audience	12+
Player Count	3–8 per match (hard minimum of 3)
Monetization	Ad revenue + cosmetic-only SD\$ economy — zero pay-to-win
Tech Stack	React.js, Node.js, Socket.io, Supabase, Cloudinary
Hosting	Vercel (frontend) + Railway (backend)
Project Lead	Mughil Sankar N
Version	2.0 — Deduction Rework

2. Character & Role Assignments

Every role in a match is skinned as a Volta Chronicles character. Role assignment is always random and server-authoritative — never purchasable, never chosen by the player. This preserves the zero pay-to-win rule established for THROWN's economy.

2.1 The Mask Roles (hidden, win by surviving the vote)

Character	Role	Player Count	Ability
The Artist	Primary Mask	3–8 (always present)	Receives a decoy clue token that reads identically to a normal Investigator clue, so clue possession alone never confirms a role.
Victor (Puppeteer)	Second Mask	6–8 only	Once per match, may secretly redirect one vote after seeing the live tally trend — a subtle nudge, not a guaranteed save.

2.2 Special Investigator Roles (randomly assigned, capped per match)

1 special Investigator role is assigned at 3–5 players; 2 are assigned at 6–8 players. Remaining seats are filled by the standard Volta Agent (no ability).

Character	Role Name	Ability
Nova Reyes	Lead Investigator	Once per match, may privately check whether one other player's clue is Strong or Weak (not its content).
Mikaela (Mileb)	Protector	Once per match, may grant one player immunity from that round's elimination vote.
Kate (Madam Volta)	Mediator	Once per match, may extend the Discussion Phase timer by +20 seconds.
Tamara (Golden Star)	Anchor	Once per match, her vote counts as 2 votes toward the elimination tally.
D.M.W.	Unshakeable	Cannot be selected as the elimination target during Round 1 of the match.
Boss	Auditor	Sees the numeric vote-count breakdown (not who voted for whom) moments before reveal.
Vanilla	Lucky Charm	If voted out and revealed as an Investigator, all remaining Investigators receive a small bonus.

2.3 Standard Role

Volta Agent — the default Investigator skin with no special ability. Fills all remaining seats not assigned to a Mask or special role. Every match therefore has at least one standard Volta Agent regardless of lobby size.

3. Entry Gate & Age Verification

Carried over from v1.0 unchanged.

- User lands on site → Age gate ('Are you 18 or older?') → No: blocked page, no further access → Yes: Rules & Regulations page (full scroll required) → Accept → Auth page
- Key rules: no real names/phone numbers in usernames or chat, wordlist-filtered conduct, report system, SDT reserves right to ban

4. Authentication & Account System

Carried over from v1.0 unchanged, since account/profile needs are identical regardless of core game loop.

Mode	Features	Limitations
Guest	Temp username, random matchmaking, room code join	No stats saved, no friends, no leaderboard placement, session only
Account	Full profile, stats, friends, leaderboard, SD\$, DP collection	Requires signup (Google / Apple / Email+Password or OTP)

- Username: letters only as first character, max 16 chars, underscore only special char
- Public profile: username, equipped DP, bio, stats (matches won, correct-vote rate, times caught as Mask), Loser Tag if active, DP collection; SD\$ balance stays private
- Multi-device: allowed, but active gameplay locks to one device
- Account deletion: must sell/scrap all DPs first; in-progress matches auto-forfeit with Loser Tag applied if applicable

5. Match & Round System

5.1 Finding a Match

Method	How It Works
Random Matchmaking	Click Find Match — random pairing into a 3–8 player lobby, cancel anytime before match starts
Create Room	Host generates SDT-XXXX room code; host sets player cap (3–8) and match length (Best of 5 / Best of 10)
Join Room	Enter room code — anyone with the code can join up to the host's player cap

5.2 Pre-Match Screen

All players see each lobby member's profile (username, DP, active Loser Tag, basic stats) before confirming. Any player can back out with no penalty. All must confirm to proceed.

5.3 Edition Vote

Scenario	Outcome
All players vote the same edition	That edition is played
Players split votes	Random pick weighted among the voted editions
Nobody votes	Host's default edition (Core) is played

5.4 Round Structure (repeats each round until match ends)

Phase	Duration	What Happens
1. Assignment	5 sec	Server secretly assigns roles + clue tokens. Never sent to other clients.
2. Discussion	45–75 sec (scales with player count)	Players talk, share/withhold clues, bluff. Mediator may extend +20s.
3. Vote	20 sec	All players vote simultaneously; hidden until timer ends, then revealed together.
4. Tie Resolution	Instant	Ties broken by random server-side tiebreak among tied players.
5. Reveal & Elimination	Instant	Voted-out player's role revealed. Mask eliminated, or Investigator removed from voting (clue history stays visible).
6. Scoreboard	Between rounds	Scores update; ad slot may trigger here (Section 8). Next round begins automatically.

5.5 Discussion Timer Scaling

Player Count	Discussion Timer
3–4	45 sec
5–6	60 sec
7–8	75 sec

5.6 Match Length & Win Conditions

- Match length fixed at lobby creation: Best of 5 or Best of 10 rounds
- Investigators win if all Mask(s) are voted out within the round limit
- Mask(s) win if the round limit is reached while at least one Mask survives
- At 6–8 players, catching only one of two Masks is not a win — both must be caught
- Final scoreboard shown at match end with instant rematch (same lobby, roles reshuffled)

6. Scoring & SD\$ Economy

SD\$ is earned entirely through gameplay and voluntary ad views — never purchased with real money. No mechanic below affects who wins a round; all purchasing/earning is cosmetic or convenience only.

6.1 Round & Match Scoring

Outcome	SD\$
Investigator correctly votes for an eliminated Mask	+150
Investigator votes incorrectly	+0
Mask survives a round without being voted out	+200
Mask is caught	+0
Winning side match bonus	+500 split evenly among winning side

6.2 Additional Earning

Action	SD\$ Earned
Scrapping a duplicate DP	Random 100–2,000
Selling a DP to a friend or auction	Seller-set price (above floor)
Voluntary Ad Box spin	Indirect — cosmetic DP or reward

6.3 Spending SD\$ (cosmetic / convenience only)

Item	Cost
Common Lootbox (cosmetic DP)	1,000 SD\$
Good Lootbox	5,000 SD\$
Legendary Lootbox	10,000 SD\$
Mythical Lootbox	50,000 SD\$
Remove Loser Tag	100 SD\$
Theme Shop items	Variable SD\$

7. Editions

Edition	Player Range	Masks	Feel
Core	3–5	1	Fast, easy to learn, classic deduction
Extended	6–8	2	Higher variance, harder Investigator coordination
Chaos Mode	6–8	2 + rotating modifier each match	Highest variance — for experienced groups

Chaos Mode modifiers are an open design track — see Section 12 (Open Questions).

8. Ad System & Monetization Rules

Hard constraint: ads and purchases never touch round outcomes, vote power, clue strength, or Mask/Investigator odds. Every player has identical mechanical power regardless of spend.

Method	Trigger	Details
Scoreboard Interstitial	Between rounds	Shown only at the Scoreboard screen — never during Discussion, Vote, or Reveal
Match-End Interstitial	After match ends	Shown on the results screen before rematch/exit
Ad Box (voluntary)	Player chooses to watch anytime	Reward: 1 free cosmetic lootbox spin (85% Common / 12% Good / 2.5% Legendary / 0.5% Mythical)

Explicitly not permitted: purchasable extra clues, vote immunity, re-rolls, or any mechanic that changes who wins a round.

9. DP Marketplace & Auction House

Carried over from v1.0, cosmetic-only (DPs are profile avatars, purely visual).

- Friends Trade: direct trade at seller-set price, above SDT floor price
- Public Auction House: 24-hour timer, escrowed bids, friends limited to 1 bid each, 5-minute snipe warning with no extension
- Floor Price System: every DP has an SDT-defined floor price to prevent scalping/deflation

10. Friends, Chat, Leaderboard, Notifications, Theme System

Carried over from v1.0 with terminology adapted to the deduction format.

10.1 Friends

- Unique Friend Code per account, mutual-accept requests, online/offline status, direct challenge if online, no friend limit

10.2 Chat

Context	Mode
Discussion Phase — randoms	Preset messages only
Discussion Phase — friends-only lobby	Free text, wordlist filtered
Post-match	Free text, wordlist filtered
Friends chat (outside match)	Free text, wordlist filtered, number-pattern blocked

10.3 Leaderboard

Board	Ranks By
SD\$ Rich	Highest SD\$ balance
Investigator Accuracy	Highest correct-vote rate
Mask Survivor	Most rounds survived as a Mask
DP Rarity	Rarest avatar owned

Real-time updates, no seasonal reset, Global + Friends tabs, guests can view but not place.

10.4 Notifications

In-app only, individually toggleable: friend online, challenge/match found, vote phase starting, round result, Loser Tag placed, friend requests, trades, marketplace activity (bid/outbid/snipe warning/auction end).

10.5 Theme System

Balatro-inspired visual identity carries over unchanged: deep dark backgrounds, neon accents, card-flip reveals. Role cards, vote UI, and reveal animations all skin to the active theme. Default theme free; additional themes purchasable with SD\$.

11. Technical Stack

Layer	Technology	Purpose
Frontend	React.js + Framer Motion	UI, animations, PWA
Backend	Node.js + Express	API, match orchestration, role/vote logic
Real-Time	Socket.io	Discussion timer, hidden vote sync, simultaneous reveal, tiebreak broadcast
Database	Supabase (PostgreSQL)	Accounts, stats, SD\$, friends, match/round/vote history
Cache / Sessions	Redis (via Supabase or Railway)	Live sessions, matchmaking queue, active rooms
Auth	Supabase Auth	Google + Apple + Email sign-in
Media	Cloudinary	Cosmetic asset delivery, DP images
Frontend Hosting	Vercel	Free tier to start
Backend Hosting	Railway	Free tier to start

12. Open Questions

- Chaos Mode modifier list — needs a dedicated brainstorm before scoping
- Voice chat built-in for Discussion Phase, or assume external call/in-person?
- Can eliminated Investigators spectate later rounds, or fully exit the session?
- Does the Oct 2 test-build deadline target Core edition only, or Core + Extended?

13. Team Structure & Work Breakdown

Mughil is Project Lead for THROWN, owning overall architecture, the core role/vote/clue game logic, sprint coordination, and final QA sign-off before each milestone. Work below is split by system so each person can build and test their area independently.

13.1 Mughil — Project Lead + Core Game Logic

- Own the overall THROWN architecture and integration between all systems below
- Build the core round-state machine: Assignment → Discussion → Vote → Tiebreak → Reveal → Scoreboard
- Implement role/clue assignment algorithm (server-authoritative randomization, special-role capping per player count)
- Implement scoring logic (Section 6.1) and win-condition checks (Section 5.6)
- Run sprint check-ins with Niranjan, Akthas, and Madur; sign off on each milestone before merge
- Own final PRD updates and QA pass before the Oct 2 test-build deadline

13.2 Niranjan — Real-Time Layer & Database

- Socket.io events: discussion timer sync, hidden vote submission, simultaneous reveal broadcast, tiebreak resolution broadcast
- Supabase schema: matches, rounds, votes, roles-per-round, clue tokens, scores
- Ensure votes stay hidden server-side until reveal (no client can see others' votes early)
- Reconnect/disconnect handling mid-round (player drops during Discussion or Vote phase)

13.3 Akthas — Backend Match Orchestration & Economy

- Lobby creation/join flow, room codes, matchmaking queue, player-cap enforcement (3–8)
- Anti-cheat safeguards: role/clue data never transmitted to non-owning clients
- SD\$ economy backend: scoring payouts, lootbox pull-rate logic, Ad Box reward validation
- Marketplace backend: auction escrow, floor price enforcement, snipe-window logic

13.4 Madur — Frontend & UI

- Role card UI (Mask/Investigator/special roles) and clue token display
- Discussion Phase UI (scaled timer, chat, Mediator extend action)
- Voting UI (hidden selection, simultaneous reveal animation, tiebreak result display)
- Scoreboard and match-end results screens
- Cosmetic shop UI (lootboxes, theme shop, Ad Box) and theme system rendering

13.5 Cross-Team Dependencies

Dependency	Blocks
Mughil's role/vote state machine	Niranjan's Socket.io events need the state machine's event contracts first
Niranjan's Supabase schema	Akthas's economy backend needs match/round tables to log payouts against
Akthas's anti-cheat role delivery	Madur's role card UI can't render real data until this is in place